

RESEARCH ARTICLE

The macroeconomy as a random forest

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Email: p.gouletcoulombe@gmail.com**Summary**

I develop the *macroeconomic random forest* (MRF), an algorithm adapting the canonical machine learning (ML) tool, to flexibly model evolving parameters in a linear macro equation. Its main output, *generalized* time-varying parameters (GTVPs), is a versatile device nesting many popular nonlinearities (threshold/switching, smooth transition, and structural breaks/change) and allowing for sophisticated new ones. The approach delivers clear forecasting gains over numerous alternatives, predicts the 2008 drastic rise in unemployment, and performs well for inflation. Unlike most ML-based methods, MRF is directly interpretable—via its GTVPs. For instance, the successful unemployment forecast is due to the influence of forward-looking variables (e.g., term spreads and housing starts) nearly doubling before every recession. Interestingly, the Phillips curve has indeed flattened, *and* its might is highly cyclical.

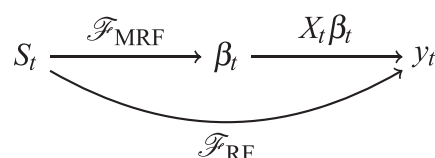
KEYWORDS

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1 | INTRODUCTION

Within the modern machine learning (ML) canon, Breiman (2001)'s random forest (RF) is outstandingly good at predicting y_t because it allows for complex nonlinearities, handles high-dimensional data, and bypasses overfitting. I redirect it to model β_t , a vector of meaningful coefficients in a macroeconomic equation.

The new algorithm, macroeconomic random forest (MRF), is summarized in the diagram below. We get a linear equation (in X_t) with time-varying coefficients following a very general law of motion. The latter is powered by a large data set (S_t) and an algorithm particularly apt with complex nonlinearities and high-dimensionality ($\mathcal{F}_{\text{MRF}}$).



The β_t process can take many forms, and $\mathcal{F}_{\text{MRF}}$ mostly dispenses with making an a priori choice out of many equally plausible alternatives. Those include threshold/switching regressions (Hansen, 2011; Sims & Zha, 2006), smooth transition (Auerbach & Gorodnichenko, 2012; Ramey & Zubairy, 2018; Teräsvirta, 1994), structural breaks (Perron, 2006), and random walk time-varying parameters (Cogley & Sargent, 2001; Primiceri, 2005). This also highlights MRF's first major advantage: It can be interpreted. Its main output, generalized time-varying parameters (GTVPs) is a versatile device nesting many popular nonlinearities. In contrast, plain RF, which can be viewed as setting $X_t = \mathbf{1}$ in $\hat{y}_t = X_t \mathcal{F}_{\text{MRF}}(S_t)$, is fully nonparametric, rather opaque, and cannot be directly linked to the vast catalog of traditional nonlinear time series models.

A second benefit is that, by striking an appealing balance of efficiency and flexibility, MRF can forecast better. Most ML algorithms are designed for large cross-sectional data sets, whereas macroeconomics is characterized by short dependent time series. If persistence (or any other linear relationship) is pervasive, important efficiency gains ensue from modeling them directly. As a result, we will see that MRF often supplants off-the-shelf ML algorithms both in simulated environments and a substantive forecasting exercise. When measured against classical nonlinear time series econometric approaches, MRF again does better but now by being less rigid about β_t 's law of motion and avoiding overfitting.

MECHANICS. The key difference when adding the M to MRF is the inclusion of a linear part within each of the tree leaves rather than just an intercept. Motivated in cross-sectional applications to improve the efficiency of nonparametric estimation (in the spirit of local linear regression), trees with linear parts have been considered in Alexander and Grimshaw (1996) and Wang and Witten (1996). Friedberg et al. (2020) expand on this by considering an ensemble of them (i.e., a forest) and focusing on the problem of treatment effect heterogeneity. Of course, the difference here is that a linear part is much more meaningful when one can look at β_t as a process of its own. The paper also introduces new devices enhancing MRF's predictive and interpretability potential. First, I propose moving average factors (MAFs) as a simple way to compress ex-ante the information contained in the lags of a regressor entering the RF part of MRF. They boost the meaningfulness of tree splits and help avoid running out of them quickly. When it comes to GTVPs themselves, I provide a regularization scheme better suited for time series which procures a desirably smoother path with respect to time. It is inspired by the random walk shrinkage of the classical TVP literature and is implemented within the tree procedure by weighted least squares. Finally, a variant of the Bayesian Bootstrap provides a tree dispersion metric that is instrumental for the interpretation of GTVPs.

RESULTS. In a forecasting application, the MRFs gains are present for almost all variables and horizons under study. For instance, the autoregressive random forest (ARRF) almost always supplant its resilient ordinary least square (OLS) counterpart. Also, an MRF where the linear part is a compact factor-augmented autoregression generates very accurate forecasts of the 2008 downturn for both GDP and the unemployment rate (UR). Inspection of resulting GTVPs reveals that they behave differently from random walk time-varying parameters (TVPs). For instance, in the UR equation, the contribution of forward-looking variables nearly doubles before every recession—including doing so *out-of-sample* (OOS) in 2008. This reinforces the view that financial indicators can rapidly capture downside risks around business cycle turning points (Adrian et al., 2019).

Inflation is subject to a variety of time variations, detection of which would be compromised by approaches lacking the generality of MRF. The long-run mean and the persistence evolved in a slow and exogenous fashion—as found in, for example, Cogley and Sargent (2001). More novel is the finding that the real activity factor's effect on the price level depends positively on the strength of well-known leading indicators, especially housing-related. Following this lead, I complete the analysis by looking at a traditional Phillips' curve (PC) specification. I report that the inflation/unemployment trade-off coefficient decreased significantly since the 1980s and also varies strongly along the business cycle. Among other things, it is extremely weak following every recession. This nuances current evidence on the flattening Phillips curve, which, by design, focused almost entirely on long-run exogenous change (Blanchard et al., 2015; Del Negro et al., 2020; Galí & Gambetti, 2019). Overall, MRF suggests that inflation can rise from a positive unemployment gap but that it goes down much more timidly from economic slack.

The appendix includes a simulation study (Appendix A1 in the supporting information). MRF is found to perform comparably well to traditional nonlinear time series models when the data generating process (DGP) matches what the latter is designed for. As expected, MRF performs better otherwise, that is, with more complex patterns. Additionally, it supplants plain RF whenever persistence is pervasive. MRF is also found to adapt nicely to various forms of time variations in a data-rich environment—whether it comes from switching dynamics, random walks, or interaction terms—and outperforms other simpler high-dimensional methods.

OUTLINE. Section 2 introduces MRF, motivates its use, considers practical aspects, and discusses relationships with available alternatives. Section 3 reports forecasting results. Section 4 analyzes various GTVPs of interest. Section 5 concludes.

2 | MRFS

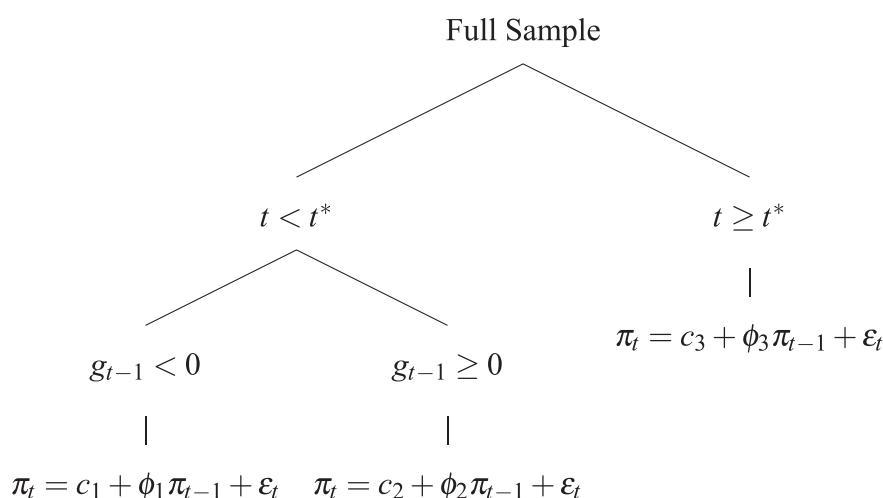
This section introduces MRF. I first motivate the use of trees as basis functions by casting standard switching structures for autoregressions as special cases. Second, I detail the MRF mechanics and how it yields GTVPs. Third, I discuss how

the approach relates to both standard RF and traditional random walk TVPs. Fourth, I discuss interpretability potential and provide a way to assess parameter uncertainty.

2.1 | Traditional macro nonlinearities as trees

RF has many qualities as a predictive algorithm, so it is a reasonable device to look into for constructing GTVPs. But there is more: Many common time series nonlinearities fit within a tree structure. Hence, it will be all the more natural to think of MRF as a generalization of previous nonlinear offerings.

Here is one example displaying how common time series nonlinearities imply a tree structure for an AR process. Let us consider the inflation process in a country where inflation targeting (IT) was implemented at a publicly known date (like in Canada). Let π_t be inflation at time t and t^* is the onset date of IT. Additionally, g_t is some measure of output gap. A plausible model is reported in the tree graph below. The story is straightforward. Inflation behaved differently before versus after IT. After IT, it is a simple AR process. Before IT, it was a switching AR process which dynamics and mean depended on the sign of the output gap.



This is one story out of many that trees can characterize. In practice, none of the above is known. The structure, the splitting variables, and the splitting points could be different. This highlights the flexibility of trees, but it also suggests that designing the “true” one from deduction is daunting—equally plausible alternatives are easily imaginable. Fortunately, algorithms can point out which trees is in better agreement with the data.

A global grid search is computationally unfeasible if either S_t is large or if we want to consider more than a few splits (examples above included 2 and 3, respectively). A natural way forward is recursive partitioning of the data set via a *greedy* algorithm (Breiman et al., 1984). A greedy algorithm optimizes functions by iteratively doing the best local update rather than directly solving for a global optimum. As a result, it is prone to high variance (Friedman et al., 2001). Hence, considering a diversified portfolio of trees appears as the most sensible route. To achieve that, it is highly effective to use Bootstrap Aggregation (*Bagging*, Breiman, 1996) of many decorrelated trees—a RF (Breiman, 2001).

2.2 | GTVPs

The general model is

$$y_t = X_t \beta_t + \epsilon_t \quad \text{with} \quad \beta_t = \mathcal{F}(S_t),$$

where S_t are the state variables governing time variation and $\mathcal{F}$ a forest. S_t is observed macroeconomic data which composition is motivated in Section 2.6 and laid out explicitly in Section 3. Importantly, including t in S_t will accommodate for stochastically evolving coefficients (discussed in detail in Section 2.5 and such a DGP is considered in the simulation study of Appendix A2 in the supporting information). X determines the *linear* model that we want to be time-varying. For instance, an ARRF—which generalizes the cases of the previous section—uses lags of y_t for X_t . The tree fitting procedure underlying *plain* RF is not adequate, as it sets $X_t = \mathbf{1}$ by default. Thus, analogously to Friedberg et al. (2020), it is

modified to

$$\min_{j \in \mathcal{J}^-, c \in \mathbb{R}} \left[\min_{\beta_1} \sum_{\{t \in |S_{j,t} \leq c\}} (y_t - X_t \beta_1)^2 + \lambda \|\beta_1\|_2 + \min_{\beta_2} \sum_{\{t \in |S_{j,t} > c\}} (y_t - X_t \beta_2)^2 + \lambda \|\beta_2\|_2 \right]. \quad (1)$$

The purpose of this problem is to find the optimal variable S_j (so, finding the best j out of the random subset of predictors indexes $\mathcal{J}^-$) to split the sample with and at which value c of that variable should we split.¹ It outputs j^* and c^* which are used to split l (the parent node) into two children nodes, l_1 and l_2 . We start with the leaf l being the full sample. Then, we perform a split according to the minimization problem, which procures us with two subsamples. Within each of these two newly created subsamples, we run (1) again. Repeating this process recursively constructs an ever-growing set of l 's which are of ever-shrinking size. Doing so until a stopping criteria is met generates a tree. Equation (1) is the sole equation characterizing the recursion that creates each underlying *tree* in MRF. What follows is the review of the three ingredients necessary to create the ensemble. It is a deliberate MRF adaptation of the textbook treatment for plain RF.

LET THE TREES RUN DEEP. Recursively splitting β_0 into β_1 and β_2 eventually leads to β_t . However, β_t , by construction, has very little company within its terminal node/leaf. As result, a single tree has low bias but also very high variance for β_t . When fitting a single tree, the (early) stopping point must be tuned to avoid overfitting. However, this is not necessary when a sufficiently diversified ensemble of trees is considered. Originally, Breiman (2001) himself provided a bound on the generalization error that grows with the correlation between trees. In Goulet Coulombe (2020b), I go further by showing that RF's OOS prediction is equivalent to the optimally “stopped” or “pruned” one, provided sufficiently diversified trees. The desirable property is attributed to the peculiar behavior of “randomized greedy algorithms”, which are often overlooked as mere computational necessities. Those insights are of even greater use when it comes to time series since dependence and structural change pose challenges to hyperparameter tuning. Given a large enough B , a reasonable `mtree` (see “decorrelation” below on this), and standard subsampling rate, we can be confident that the out-of-bag prediction and β_t 's exclude fitted noise. In our specific context, it means the sample will not be oversplit, and we are not going to see time variation when it is not there. Naturally, the dispersion metric proposed in Section 2.7 will also help in that regard. The property will be illustrated in the simulation study (Appendix A2 in the supporting information).

(M)RF prediction is the *simple average* from those of its single trees. Same goes for β_t . RF is a clever diversification scheme which generates sufficient randomization for that average to inherit the above properties. To achieve that, it mixes elements of resampling and model averaging: bagging and decorrelated trees.

BAGGING. Each tree is “grown” on a bootstrapped sample (or a random subsample) (Breiman, 1996). When the base learner is highly nonlinear in observation and/or unstable, gains from Bagging can be large (Breiman, 1996). Nonparametric (or “pairs” MacKinnon, 2006) bootstrap is being used—that is, we are *not* shuffling residuals. Rather, we are randomly selecting many observation triples $[y_t, X_t, S_t]$ (or pairs $[y_t, S_t]$ for Plain RF) and then fit a tree on them. To deal with the dependence inherent to time series data and other reasons detailed in Section 2.7, a slightly more sophisticated bootstrapping/subsampling procedure (involving blocks) will be used for MRF.

DECORRELATION. The second ingredient, proposed in Breiman (2001), is to consider “decorrelated” trees. RF is an average of many trees, and any averaging scheme reduces variance at a much faster rate if its components are uncorrelated. In our context, this is obtained by growing trees semistochastically. In Equation (1), this is made operational by using $\mathcal{J}^- \subset \mathcal{J}$ rather than $\mathcal{J}$. In words, this means that at each step of the recursion, a different subsample of regressors is drawn to constitute candidates for the split. This prevents the greedy algorithm (which, as we know, only “thinks” locally) to always embark on the same optimization route. As a result, trees are further diversified and computing time reduced. The fraction of randomly selected predictors is a tuning parameter typically referred to as `mtree` in the literature (and all software), with a default value of $\frac{1}{3}$ for regression settings. This, other algorithmic parameter settings, and some practical aspects are discussed in Appendix A6 in the supporting information.

Plain RF has many qualities readily transferable to MRF. It is easy to implement and to tune. That is, it has few tuning parameters that are usually of little importance to the overall performance—robustness. It is relatively immune to the adverse effects of including many irrelevant features (Friedman et al., 2001). Given the standard ratio of regressors to observations in macro data, this is a nonnegligible advantage. Furthermore, with a sufficiently high `mtree`, it can adapt

¹Note that, unlike Friedberg et al. (2020), S_t and X_t will differ, which is natural when motivated from a TVP perspective (but not so much from local linear regression one). Forcing their equivalence is not feasible nor desirable in a macro environment. However, what is desirable and used throughout is $X_t \subset S_t$, which allows, if need be, for nonparametric correction of the linear model based on X_t 's themselves.

nicely to sparsity and discard useless predictors (Olson & Wyner, 2018). Finally, its vanilla version already shows good macroeconomic forecasting performance (Chen et al., 2019; Goulet Coulombe et al., 2022; Medeiros et al., 2021).

2.3 | Random walk regularization

Equation (1) uses ridge shrinkage which implies that each time-varying coefficient is implicitly shrunk to 0 at every point in time. λ and the prior it entails can exert a significant influence. For instance, if a process is highly persistent, shrinking the first lag heavily to 0 could incur serious bias. If X_t is low-dimensional, using OLS coefficients as prior means is potential fix. Nonetheless, the traditional regularization employed in macro is rather the random walk ($\beta_t = \beta_{t-1} + u_t$). Thus, it is desirable to transform (1) so that it implements the prior that coefficients evolve semismoothly, which is just shrinking β_t to be in the neighborhood of β_{t-1} and β_{t+1} rather than 0. This is in line with the view that economic states last for at least a few consecutive periods. Note that unlike traditional TVP methods which rely extensively on smoothness regularization—as it is the sole regularizer, MRF makes only a very mild use of it to get rid of high-frequency noise that may be left in β_t . The main benefit is to facilitate the interpretation of resulting GTVPs.

I implement the desired regularization by taking the “rolling-window view” of time-varying parameters, which has been exploited recently to estimate large TVP-VARs (Giraitis et al., 2018; Petrova, 2019). That is, the tree, instead of solving a plethora of small ridge problems, will rather solve many weighted least squares (WLS) problems which include close-by observations. The latter are in the neighborhood (in time) of observations within current leaf. They are included in estimation with a smaller weight.

For simplicity and to keep computational demand low, the kernel used by WLS is rather rudimentary: It is a symmetric five-step Olympic podium. Essentially, it is a reparametrized² truncated (on both sides) exponential window around t . It will be formalized below for the application at hand which involves, among other things, dealing with leaves. But first, informally, the kernel puts a weight of 1 on observation t , a weight of $\zeta < 1$ for observations $t - 1$ and $t + 1$ and a weight of ζ^2 for observations $t - 2$ and $t + 2$. Since some specific t 's will come up many times (e.g., if both observations t and $t + 1$ are within the same leaf, podiums overlap), I take the maximal weight allocated to t as the final weight $w(t; \zeta)$.

Formally, define L_{-1} as the “lagged” version of leaf l . In other words, L_{-1} is a set containing each observation from l , with all of them lagged one step. L_{+1} is the “forwarded” version. L_{-2} and L_{+2} are two-step equivalents. For a given candidate subsample l , the podium is

$$w(t; \zeta) = \begin{cases} 1, & \text{if } t \in l \\ \zeta, & \text{if } t \in (L_{+1} \cup L_{-1})/l \\ \zeta^2, & \text{if } t \in (L_{+2} \cup L_{-2})/(l \cup (L_{+1} \cup L_{-1})) \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

where $\zeta < 1$, a tuning parameter guiding the level of time-smoothing. Then, it is only a matter of how to include those additional (but down weighted) observations in the tree search procedure. The usual candidate splitting sets

$$l_1(j, c) \equiv \{t \in l | S_{j,t} \leq c\} \quad \text{and} \quad l_2(j, c) \equiv \{t \in l | S_{j,t} > c\}$$

are expanded to include all observations of relevance to the podium

$$\text{for } i = 1, 2 : l_i^{RW}(j, c) \equiv l_i(j, c) \cup l_i(j, c)_{-1} \cup l_i(j, c)_{+1} \cup l_i(j, c)_{-2} \cup l_i(j, c)_{+2}.$$

The splitting rule becomes

$$\min_{j \in J^-, c \in \mathbb{R}} \left[\min_{\beta_1} \sum_{t \in l_1^{RW}(j, c)} w(t; \zeta)(y_t - X_t \beta_1)^2 + \lambda \|\beta_1\|_2 + \min_{\beta_2} \sum_{t \in l_2^{RW}(j, c)} w(t; \zeta)(y_t - X_t \beta_2)^2 + \lambda \|\beta_2\|_2 \right]. \quad (3)$$

Note that the ridge penalty is conserved, so the final model has in fact two sources of regularization. With $\zeta \rightarrow 0$, we are heading back to pure ridge.

²This is done so the tuning parameter makes intuitive time series sense as exponential decay such as what one would get for the implied linear kernel of an AR(1) process (i.e., its autocovariances). Thus, one can easily think of default values for ζ .

2.4 | Relationship to standard random forest

The standard RF is MRF with $X_t = 1$, $\lambda = 0$ and $\zeta = 0$. In words, the only regressor is a constant, and there is no within-leaf shrinkage. From a ML standpoint, why should we prefer the partially linear MRF to the fully nonparametric RF? One reason is statistical efficiency. The other is potential for interpretation.

THE HARDSHIP OF SMOOTH RELATIONSHIPS. In finite samples, plain RF can have a hard time learning smooth relationships—like a AR(1) process. For prediction purposes, estimating $y_t = \phi y_{t-1} + \varepsilon_t$ by OLS implies a single parameter. However, approximating the same relationship with a tree (or an ensemble of them) is far more consuming in terms of degrees of freedom. To get close to the straight line once parsimoniously parametrized by ϕ , we now need a succession of many step functions. With short time series, one runs out of degrees of freedom before capturing adequately ϕ . For more complex DGPs, the mechanical consequence is that RF will waste many valuable splits on capturing the linear part before it gets to focus more subtle nonlinear phenomena. One symptom is AR terms being flagged as really important by typical RF variable importance (VI) measures. MRF provides a workaround. Modeling the linear part concisely leaves more room to estimate the nonlinear one. By its more strategic budgeting of degrees of freedom, the resulting partially linear model could be, in fact, more nonlinear than the fully nonparametric one.

This paper is not the first to recognize the potential need for a linear part in tree-based models. For instance, both Alexander and Grimshaw (1996) and Wang and Witten (1996) proposed linear regressions within a leaf of a tree. Friedberg et al. (2020) proposed to improve the nonparametric estimation of treatment effect heterogeneity by combining those ideas developed for trees into a forest. This paper is the first to exploit the link between this strand of work and the semipiternal search for the “true” state dependence in empirical macroeconomic models. In contemporaneous work, Farrell et al. (2020) highlight the potential for interpretability of partially linear neural networks in a heterogeneous treatment effect context.

A NOTE ON INTERPRETABILITY. The interpretation of ML outputs is now a field of its own (Molnar, 2019). RF is widely regarded as a black box model which needs to be interpreted using an external device. Indeed, it usually averages over 100 trees of substantial depth, which makes individual inspection impossible. MRFs partially circumvent the problem by providing time series β_t which can be examined and have a meaning as time-varying parameters for the linear model. Thus, whatever one may do with TVPs, it can be done with GTVPs. There are also some new avenues. For instance, VI measures usually deployed to dissect RF's prediction can be used to inspect what is driving β_t 's. Those will be used in Section 4.3. Another popular approach to dissect a standard RF is to use interpretable surrogate tree models to partially replicate the black box model's fit (Molnar, 2019). The partial linearity of MRF facilitates such an exercise. The linear part in MRF splits the conditional mean into different pieces ($X_{t,k}\beta_{t,k}$), and each time series $\beta_{t,k}$ can be dissected with its own surrogate model.

2.5 | Relationship to random walk time-varying parameters

GTVPs have many advantages over classical TVPs. While it is known that any nonlinear model can be approximated by a linear one with TVPs (Granger, 2008), nothing is said about how efficient that estimation is going to be. As it turns out, efficiency crucially matters in a macro context, and random walk TVPs can be quite inefficient (Aruoba et al., 2017). For example, if the true β_t follows a recurrent switching mechanism, random walk parameters already have two strikes against them. Some dimensionality reduction techniques can help, but nothing in that paradigm can come close to the parsimony of simply interacting X_t with relevant variables. In contrast, MRF considers all time-variation options and chooses the “obvious thing”, which may or may not be splitting on t . Also, it is absolutely possible that the resulting $\mathcal{F}$ pools both latent and observable time variation.

Even though MRF is remarkably flexible, its variance remains low thanks to the diversified portfolio of trees. The variance of classical TVPs can be controlled by cross-validation (Goulet Coulombe, 2020a) or via an elaborate hierarchical prior (Amir-Ahmadi et al., 2020). A number of applications opt for a “manual” approach (D'Agostino et al., 2013). However, it is understood that no tuning, however, careful it may be, can overcome the hardship of fitting random walks when the true β_t 's look nothing like it.

Econometrically, one way to more formally connect this paradigm to recent work on TVPs is to adopt the view that RFs are adaptive kernel estimators (Lin & Jeon, 2006). That is, the tree ensemble is a machine generating kernel weights. Once those are obtained, estimation amounts to WLS problem with a ridge penalty. By running (1) recursively, one obtains terminal nodes/leaves $L_b()$ to construct kernel weights

$$\alpha_t(x_0) = \frac{1}{B} \sum_{b=1}^B \frac{1\{X_t \in L_b(x_0)\}}{|L_b(x_0)|} \text{ to use in } \forall t : \operatorname{argmin}_{\beta_t} \left\{ \sum_{\tau=1}^T \alpha_t(s_\tau) (Y_\tau - X_\tau \beta_\tau)^2 + \lambda \|\beta_t\|_2 \right\}. \quad (4)$$

As utilized in Goulet Coulombe (2020a)'s two-step dual ridge regression approach, standard random walk TVPs are in fact a smoothing splines problem. Giraitis et al. (2018) drop the random walk altogether and proposed to use kernels directly and show that such an approach can estimate consistently stochastically evolving coefficients. The common kernel view helps linking GTVPs to traditional TVPs. In the latter case, the only variable entering the kernel is t . In other words, only proximity in time is considered for the clustering of observations. This makes the seemingly flexible estimator in fact quite restrictive—and dependent on its inherent smoothness prior. Moreover, standard kernel methods are known to break down even in medium dimensions (Friedberg et al., 2020). Therefore, augmenting t —the sole variable in the kernel (implicit of explicit) of traditional TVP methods—with additional regressors is not promising. In contrast, RF can handle computationally large amount of regressors and is helped statistically by suffering less from the presence of useless predictors (Olson & Wyner, 2018).

2.6 | Engineering S_t

We need not choose between a sparse or a dense representation of regressors for S_t as we can supply RF all relevant linear combinations of original inputs and let it select what appears relevant. I introduce here a simple transformation that is a great addition to S_t with the raw data and traditional principal component analysis (PCA)-based factors.

LAG POLYNOMIALS. From a predictive standpoint, residuals autocorrelation implies that there is forecasting power left on the table. To get rid of it, many lags might be necessary. In multivariate contexts (like that of a VAR), doing so quickly pushes the model to overfit. A standard solution is Bayesian estimation and the use of priors specially designed for blocks of lags structures. Outside of the VAR paradigm, there is an older literature estimating restricted/regularized lag polynomials in autoregressive distributed lags (ARDL) models (Almon, 1965). (M)RF experiences an analogous situation. A tree may waste many splits trying to efficiently extract information out of a lag polynomial: for instance, splitting on the first lag, then the seventh one, then the third one. In linear parametric models, the above methods can extract the relevant information out of a lag polynomial without sacrificing many degrees of freedom. A significant roadblock to this enterprise in the RF paradigm is that there are no explicit lag polynomials to penalize. An alternative route is to exploit the insight that RF can choose for itself relevant restrictions. We just have to construct regressors that embody those and include them in S_t .

MAFs. To extract the essential information out of the lag polynomial of a specific variable, a linear transformation can do the job. Consider forming a panel of P lags of variable j :

$$X_{t,j}^{1:P} \equiv [X_{t-1,j} \dots X_{t-P,j}] .$$

We want to form weighted averages of the P lags so that it summarizes most efficiently the temporal information of the feature indexed by j . The weighted averages with that property will be the first few factors (extracted by PCA) of $X_{t,j}^{1:P}$. This can be seen as the time-dimension analog to the traditional cross-sectional factors. The latter are defined such as to maximize their capacity to replicate the cross-sectional distribution of $X_{t,j}$ fixing t , while the MAFs proposed here seek to sparsely represent the temporal distribution of $X_{t,j}$ for a fixed j in a lower-dimensional space. By doing so, our goal to summarize the information of $X_{t,j}^{1:P}$ without modifying the RF algorithm is achieved by utilizing MAFs which compress information ex-ante. RF's job is then to select the relevant linear combinations among S_t so to maximize the fit. Finally, MAFs facilitate interpretation as these are moderately sophisticated averages of a single time series and can be viewed as a smooth index for a tangible economic indicator.

2.7 | Quantifying uncertainty of β_t 's estimates

Taddy et al. (2015, 2016) interpret RF's prediction as the posterior mean of a tree functional $\mathcal{T}$ (the splitting algorithm) obtained by an approximate Rubin (1981)'s Bayesian bootstrap. Through those lenses, each tree is a posterior draw. Seeing $\mathcal{T}$ as a Bayesian nonparametric statistic (independently of the DGP) is of even greater interest in the case of MRF. It provides uncertainty quantification for meaningful time-varying parameters β_t rather than an opaque conditional mean

function. Recently, Newton et al. (2021) have shown its validity (through first-order asymptotics) for applications ranging from regularized regression to deep learning.

As noted in Taddy et al. (2016), this tree dispersion metric is informative, easily obtainable, and can be linked to an approximate Bayesian posterior distribution. However, they do not go as far as establishing the frequentist properties of the estimator. An alternative (frequentist) inferential approach is that of Friedberg et al. (2020). However, their asymptotic argument requires estimating the linear coefficients and the kernel weights on two different subsamples. This is hard to reconcile with our goal of modeling time variation and different regimes throughout the entire sample. Furthermore, when the sample size is small, splitting the sample in such a way carries binding limitations on the complexity of the estimated function. Thus, the tree Bayesian bootstrap view appears as the most practical approach for the applications at hand in this paper.

While the Bayesian Bootstrap desirably does not assume many things about the data, it yet makes the assumption that $Z_t = [y_t \ X_t \ S_t]$ is an *iid* random variable. In a time series context, it is more appropriate to use blocks of observations. Thereby, I propose a block extension to make Taddy et al. (2015)'s convenient approach amenable to this paper's setup. Block Bayesian bootstrap (BBB) is a simple redefinition of Z so that it is plausibly *iid*. Hence, in the spirit of traditional frequentist block bootstrap (MacKinnon, 2006), blocks of a well-chosen size will be exchangeable. Thus, a new variable can be defined $Z_b \equiv [y_{b:\bar{b}} \ X_{b:\bar{b}} \ S_{b:\bar{b}}]$. There will be a total of $\mathfrak{B} = T/\text{block size}$ fixed and nonoverlapping blocks. Under strong mixing conditions, $\tilde{Z}_b = \text{vec}(Z_b)$ are *iid*, for a properly chosen block length (Politis et al., 2001). Analogously to Taddy et al. (2015), block subsampling is preferred to BBB in implementations since it is faster and gives nearly identical results. Details of BB and BBB, as well as a discussion of the treatment of potential heteroscedascity, are all available in Appendix A4 in the supporting information.

3 | MACROECONOMIC FORECASTING

I conduct a pseudo-OOS forecasting experiment using the quarterly frequency dataset FRED-QD (McCracken & Ng, 2020) which contains 248 US macroeconomic and financial aggregates observed from 1960Q1. The forecasting targets are real GDP, UR, CPI inflation (INF), 1-year treasury constant maturity rate (IR), and the difference between 10-year treasury constant maturity rate and federal funds rate (SPREAD). These series are representative macroeconomic indicators of the US economy which is based on Goulet Coulombe et al. (2022) exercise for many ML models. The series transformations to induce stationarity for predictors are indicated in McCracken and Ng (2020). For forecasting targets, GDP, UR, CPI, and IR are considered $I(1)$ and are first-differenced. For the first two, the natural logarithm is applied before differencing. SPREAD is kept in “levels”. Forecasting horizons are 1, 2, 4, 6, and 8 quarters. The pseudo-OOS period starts in 2003Q1 and ends in 2014Q4. I use expanding window estimation from 1961Q3. Models are estimated (and tuned, when applicable) every 2 years. For all models except self-exciting threshold AR (SETAR) and smooth transition threshold AR (STAR), I use direct forecasts, meaning that $\hat{y}_{t+h}$ is obtained by fitting the model directly to y_{t+h} rather than iterating one-step ahead forecasts. SETAR and STAR iterated forecasts are calculated using the block bootstrap method (Clements & Smith, 1997). Following standard practice, the quality of point forecasts is evaluated using the root mean square prediction error (MSPE). For the OOS forecasted values at time t of variable v made h steps ahead, I compute

$$RMSE_{v,h,m} = \sqrt{\frac{1}{\#OOS} \sum_{t \in OOS} (y_t^v - \hat{y}_{t-h}^{v,h,m})^2}.$$

The Diebold and Mariano (2002) (DM) test procedure is used to compare the predictive accuracy of each model against the reference AR(4) model.

It has been argued in Section 2.6 that feature engineering matters. S_t , the set of variables from which RF can select, is motivated by such concerns. Its exact composition is detailed in Table 1. Among other things, it includes both cross-sectional and MAFs, which are compressing information along their respective dimensions.

MODELS. To better understand where the gains from MRF are coming from, I include models that use different subsets of ideas developed in earlier sections. Those are summarized in Table 2. The competitive data-rich models are in the benchmarks group. Nonlinear time series models are also included as they share an obvious familiarity with ARRF. “Tiny” versions of both ARRF and RF are considered to gauge the effect from only having access only to a small S_t —this could be the case for many non-US applications. Conversely, this helps quantify how a data-rich environment contributes to the

TABLE 1 Composition of S_t .

What	Why	How
Eight lags of y_t	Endogenous SETAR-like dynamics	–
t	Exogenous “structural” change/breaks	–
Two lags of each variable in FRED-QD	Fast switching behavior	–
Eight lags of 5 traditional factors F	Compress cross-sectional information	Usual PCA
Two MAFs for each variable j	Compress lag polynomial information	PCA on eight lags of j

Abbreviations: MAF, moving average factor; SETAR, self-exciting threshold AR.

TABLE 2 Forecasting models.

Name	Acronym	Linear part (X_t^m)	Nonlinear part
Autoregression	AR	$[1, y_{t-1:4}]$	$\emptyset$
Factor-augmented autoregression	FA-AR	$[1, y_{t-1:4}, F_{1,t-1:2}, F_{2,t-1:2}]$	$\emptyset$
Plain random forest	RF	$\emptyset$	Raw data ^a
Low-dimensional plain RF	Tiny RF	$\emptyset$	$[y_{t-1:8}, t]$
Deep neural network ^b	NN	$\emptyset$	S_t
Plain RF but using S_t	RF-MAF	$\emptyset$	S_t
RF-MAF on decorrelated y_t	AR+RF	Filter y_t first with an AR(2), then RF	S_t
Autoregressive random forest	ARRF	$[1, y_{t-1:2}]$	S_t
Low-dimensional autoregressive RF	Tiny ARRF	$[1, y_{t-1:2}]$	$[y_{t-1:8}, t]$
Factor-augmented autoregressive RF	FA-ARRF	$[1, y_{t-1:2}, F_{1,t-1}, F_{2,t-1}]$	S_t
Vector autoregressive RF ^c	VARRF	$[1, y_{t-1:2}, GDP_{t-1}, IR_{t-1}, INF_{t-1}]$	S_t
Self-exciting threshold AR	SETAR	$[1, y_{t-1:2}]$	$\emptyset$
Smooth transition AR ^d	STAR	$[1, y_{t-1:2}]$	$\emptyset$
10 years rolling-window AR	RW-AR	$[1, y_{t-1:2}]$	$\emptyset$
Time-varying parameters AR ^e	TV-AR	$[1, y_{t-1:2}]$	$\emptyset$
Time-varying parameters FA-AR ^f	TV-FAAR	$[1, y_{t-1:4}, F_{1,t-1:2}, F_{2,t-1:2}]$	$\emptyset$
LASSO using S_t	LASSO-MAF	S_t	$\emptyset$
Ridge using S_t	Ridge-MAF	S_t	$\emptyset$

Note: Models are classified in three categories: benchmarks, MRFs (and related prototypes), and misc (nonlinear time series models, other reasonable additions). The main analysis in Section 3.1 omits the third club for parsimony.

^aPrecisely, this means 8 lags of FRED-QD, after usual transformations for stationarity have been applied.

^bFollowing a similar architecture as in Goulet Coulombe (2020b), see Appendix A2 in the supporting information for details.

^cNote that the VAR appellation refers to the linear equation consisting of typical “small monetary VAR”. The model remains univariate and direct forecasts are used.

^dThe state variable is y_{t-1} , as in SETAR.

^eEstimated and tuned via the Ridge approach proposed in Goulet Coulombe (2020a).

^fNote that this is partially different from what gave the results reported in Section 3.1, where the model was re-estimated every 2 years. Here, estimation occurs once in 2007Q2.

success of ARRF versus its plain flexibility. Indeed, tiny ARRF corresponds to what is also considered in the “data-poor” simulations (Appendix A.1.1 in the supporting information) to be a generalization of SETAR/STAR and related models.

Here are some remarks motivating some inclusions and specifications choices. To assess the marginal effects of MAFs alone, Lasso, ridge, and RF are considered using S_t —those are known to handle high-dimensional feature spaces.³ When it comes to FA-ARRF, I opt for a parsimonious linear specification including one lag of the first two factors. First, concise models make interpretation easier. Second, considering compact linear specifications within MRF is usually the better strategy. Parameters (including the intercept) are all RFs in their own right and can palliate to the omission of marginally important features, if need be. Consequently, it is desirable to fix a humble linear part and let β_t 's take care of the rest. Finally, as discussed in McCracken and Ng (2020), the first factor mostly loads on real activity variables, while the second is a composite of forward-looking indicators like term spreads, permits, and inventories. They are baptized and interpreted accordingly.

³A version of Lasso and ridge with interactions terms of S_t and X_t was considered, as it is natural in this environment but was always inferior to plain RF using S_t , which internally creates its own interaction terms.

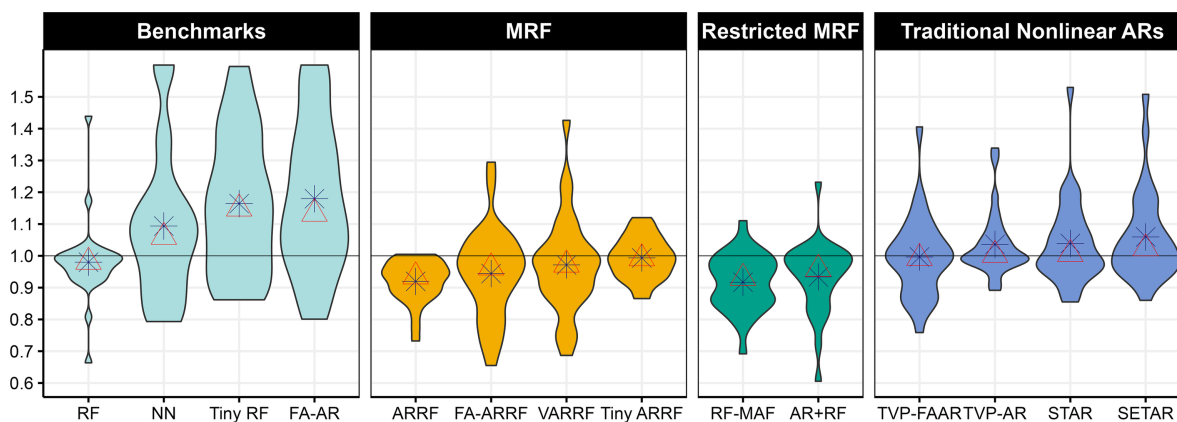


FIGURE 1 The distribution of $RMSE_{v,h,m}/RMSE_{v,h,AR}$. The star is the mean, and the triangle is the median.

3.1 | Main quarterly frequency results

Violin plots are used throughout to summarize dense RMSEs tables (like Table A2 in the supporting information). I report the distribution of $RMSE_{v,h,m}/RMSE_{v,h,AR}$. This is informative about the overall ranking and versatility of considered models. Of course, being ranked first does not imply being the best model for every h and v . Rather, it means that it performs better on average, over all targets.

Here are interesting observations from Figure 1. Clearly, MRFs deliver important gains over both the AR and FA-AR benchmarks (the latter is second to last). ARRF has a noticeably small mass above the 1 line. In other words, there are no targets for which ARRF does significantly worse than its OLS counterpart, which makes it atypically adaptable among nonlinear autoregressions. A look at Table A2 in the supporting information confirms this observation also extends to FA-ARRF versus FA-AR. We also see that simpler time variation as in the ridge-based TVP-FAAR from Goulet Coulombe (2020a) can provide appreciable ameliorations at very long horizons, but only FA-ARRF thrives at horizons one, two, and four quarters. The simplification AR+RF (run plain AR then RF, a form of additive model) ranks third with a performance that is much more volatile. This suggests that imposing time-invariant dynamics can sometimes help (see one example in Figure 3) but can also be highly detrimental (as reported for inflation). Of course, that we do not know ex-ante, and it is why AR+RF does not inherit ARRF's “off-the-shelf” quality.

MAFs are useful: RF-MAF does much better than RF which uses the raw data. The latter only exhibits conservative gains over the benchmark. Thus, it is understood that a fraction of MRFs' forecasting gains emanates from considering more sensible transformations of time series data—and which are trivially implementable. They are also what powers the NN, which never shines in any particular way nor seems more consistent in performance than RF-MAF, except for two quarters ahead forecasts of GDP and inflation where it provides the best forecast by a small margin.

FA-ARRF provides very substantial improvements but can also fail. This is the linear part's doing: Results show that FA-AR itself is prone large gains and losses. Accordingly, it is not surprising to see FA-ARRF inherit some of these uneven properties, albeit to a much milder extent. For instance, in Table A2 in the supporting information, FA-AR is noticeably worse than AR for all inflation horizons, while FA-ARRF beats AR for all of them. This phenomenon is well summarized by FA-AR being second to the last *overall*, well behind FA-ARRF. Allowing for basic random walk-based time variation (TVP-FAAR) also helps greatly in mitigating some of losses but to a lesser extent. VARRF has a behavior similar to that of FA-ARRF but with less highly noticeable gains.

Does a large S_t pay off? Most of the time, yes. It is worth re-emphasizing that restricting S_t restricts the space of time variations possibilities as well as the potential for trees diversification. Nonetheless, if the restrictions are “true”, gains are possible. This is reported to be a rare occurrence, with $ARRF > \text{Tiny ARRF}$ (and $RF > \text{Tiny RF}$) for almost any target. Thus, we can safely conclude that a rich S_t is desirable, with $\mathcal{F}$ being tasked with selection of relevant items. As discussed in earlier sections, ARRF connects to the wider family of nonlinear autoregressive models. It clearly does better on average than SETAR and smooth transition TAR. This advantage is attributable to both a more flexible law of motion and a large S_t . Tiny ARRF is better than the “Traditional Nonlinear ARs” group, while ARRF is *much* better.

REAL ACTIVITY TARGETS. Figure 2 reports results for UR. FA-ARRF dominates strongly. Table A2 in the supporting information confirms that it is the best model for all horizons but the last one (eight quarters ahead, where the encompassed RF-MAF is the best). Clearly, at a horizon of one quarter, the preferred model successfully predicts the drastic

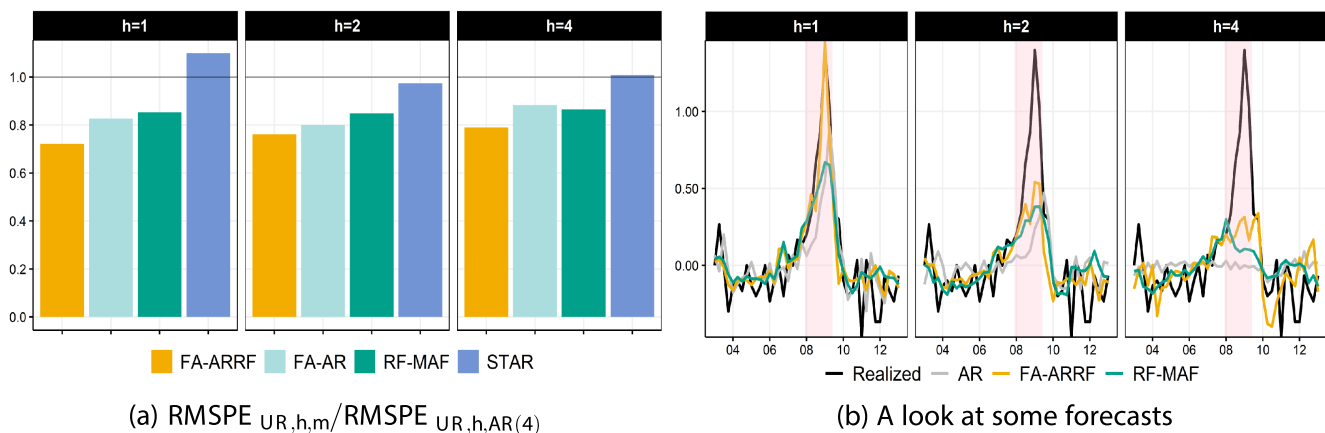


FIGURE 2 Zooming on best model within each group for unemployment rate (UR) (change).

rise in unemployment during the Great Recession. Rather than responding with a lag to negative shocks (which is what we observe from AR and ARRF), the model visibly predicts them. As a result, improvements in RMSE are between 25% and 30% over AR for all horizons. Specifically, predicting UR (change) with FA-ARRF at $h = 1$ yields an unusually high OOS R^2 of about 80%. The nearly perfect overlap of the yellow and black lines highlight the absence of a one-step ahead shock around 2008. Note that FA-AR and smooth transition threshold AR forecasts are omitted from Figure 2b to enhance visibility. Smooth transition threshold AR forecasts are either similar or worse than the benchmark (as often found for nonlinear time series models). FA-AR forecasts at $h = 1$ follows a proactive pattern similar to the yellow line but with a 1 quarter to 2 quarters delay—hence, the inferior results.

For $h = 2$, the quantitative rise is nowhere near the realized one, but it reveals 6 months ahead the arrival of a significant economic downturn. Additionally, ARRF and FA-ARRF both flag 1 year ahead the arrival of a rise in unemployment, which is a quality shared by very few models. The barplot in Figure 2 (and Table A1 in the supporting information) provides a natural decomposition of FA-ARRF's gains. Adding the MAFs to an otherwise plain RF procures an improvement of roughly 15% across all horizons (RF-MAF > RF, in Table A2 in the supporting information). The linear FA-AR part and the rest of algorithmic modifications discussed in Section 2 provide an additional reduction of 10% to 15% depending on the forecasted horizon (FA-ARRF > RF-MAF and FA-ARRF > FA-AR). It is noteworthy that good results for $h = 1$ (and other shorter horizons where the forecast can be something else than some long-run mean) are mechanically close to impossible with a plain RF since it cannot extrapolate—that is, predict values of y_t that did not occur in-sample. In contrast, this is absolutely feasible within MRF thanks to the linear part.

GDP growth is alleged to have a lower signal-to-noise ratio (Aruoba et al., 2013). In Figure A10 in the supporting information, FA-ARRF exhibits a bit less than a 20% drop in RMSE over the AR and nicely grasp the 2008 drop one quarter ahead. However, FA-ARRF performance does not stand apart as much as it did for UR. One reason can be traced visually to predicting higher postrecession growth than its competitors. Finally, RF-MAF closing in on ARRF will be investigated on its own in Section 4.2.1. In short, this occurs because once the time-varying intercept is flexibly modeled by RF, there is very little room left for autoregressive behavior (at the quarterly frequency).

4 | ANALYSIS

Based on forecasting results, I concentrate on analyzing the GTVPs of FA-ARRF's (i.e., the MRF with a factor-augmented autoregressive linear part). Some of it requires many “intermediary” calculations, like VI metrics, and the figures reported can be found in Appendix A8 in the supporting information.

4.1 | Forecast anatomy

β_t 's characterize completely MRF's forecasts. Thus, we can investigate GTVPs to understand results from the previous section. The FA-ARRF forecasting equation is

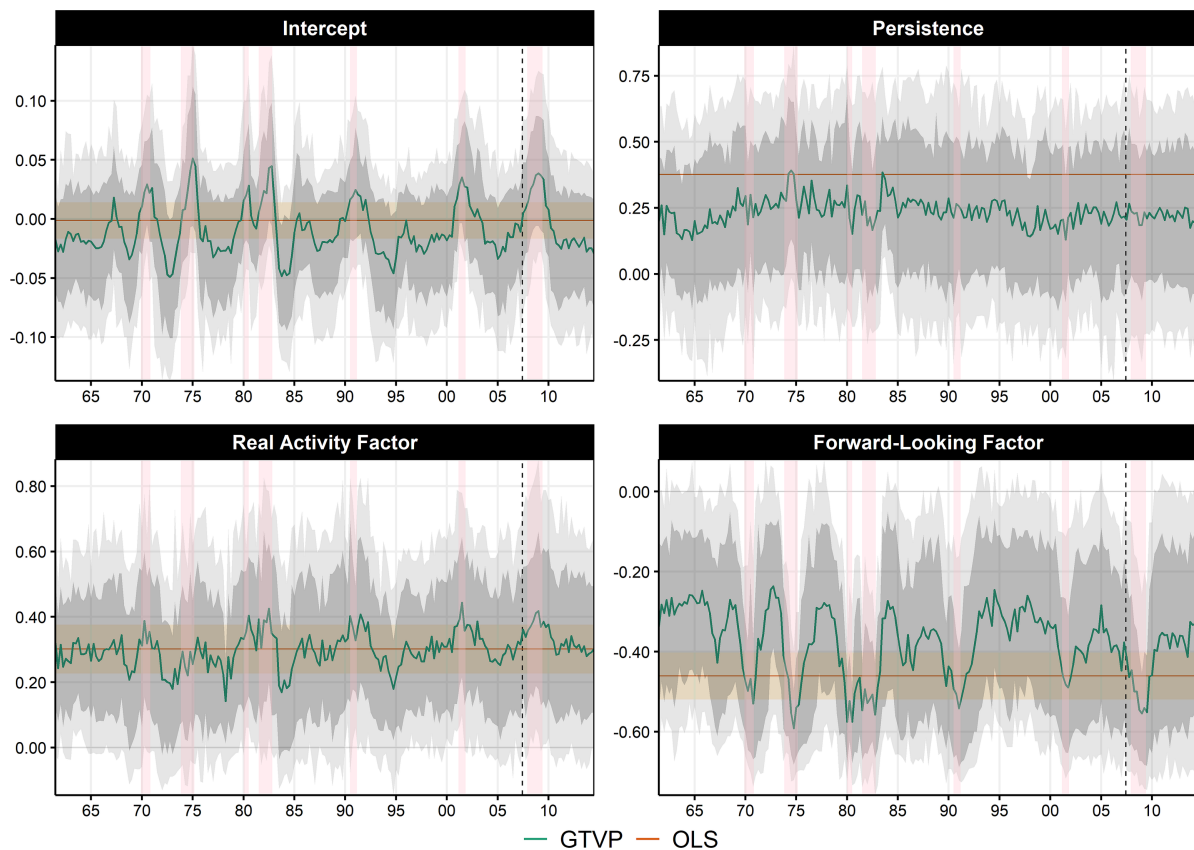


FIGURE 3 Generalized time-varying parameters (GTVPs) of the one quarter ahead unemployment rate (UR) forecast. Persistence is defined as $\phi_{1,t} + \phi_{2,t}$. The gray bands are the 32%–68% and 10%–90% quantiles of the tree distribution. The pale orange region is the OLS coefficient $\pm$ one standard error. The vertical dotted blue line is the end of the training sample. Pink shading corresponds to NBER recessions.

$$y_{t+h} = \mu_t + \phi_{1,t}y_t + \phi_{2,t}y_{t-1} + \gamma_{1,t}F_{1,t} + \gamma_{2,t}F_{2,t} + u_{t+h}$$

and naturally $\beta_t = [\mu_t \ \phi_{1,t} \ \phi_{2,t} \ \gamma_{1,t} \ \gamma_{2,t}]$. To avoid overfitting, $\hat{\beta}_t$'s are (as in Section A2 in the supporting information) the mean over draws that did not include observations $t - 4$ to $t + 4$ (a 2-year block) in the tree fitting process. Intuitively, this mimics in-sample the real OOS experiment that starts here in 2007Q2.⁴

Figure 3 displays GTVPs underlying the successful one-step ahead UR *change* forecast. The intercept clearly alternates between at least two regimes, and the “increasing UR” one is in effect circa 2008. In levels, this translates to UR alternating between a positive trend and a negative (albeit small) trend. Overall persistence is strikingly time-invariant and marginally smaller than for OLS estimates. The effect of F_1 , the real activity factor, is generally within OLS confidence intervals, suggesting that while $\gamma_{1,t}$ almost doubles around recessions, this is subject to great uncertainty.

What is less uncertain, however, is the magnified contribution of the forward-looking factor F_2 during recessionary episodes, which stands out as the key difference with OLS. $\gamma_{2,t}$ smooth switching behavior can be best interpreted by remembering that F_2 is highly correlated with capacity utilization, manufacturing sector indicators, building permits, and financial indicators (like spreads) (McCracken & Ng, 2020). Many of those variables are considered “leading” indicators and have often been found to increase forecasting performance, mostly before and during recession periods (Estrella & Mishkin, 1998; Leamer, 2007). Recently, there has been renewed attention on the matter, with financial indicators highlighted as capable of capturing economic activity downside risk (Adrian et al., 2019; Delle Monache et al., 2020). This brand of nonlinearity can translate to a more active $\gamma_{2,t}$ around business cycle turning points. MRF learns that while OLS

⁴Note that this is partially different from what gave the results reported in Section 3.1, where the model was re-estimated every 2 years. Here, estimation occurs once in 2007Q2.

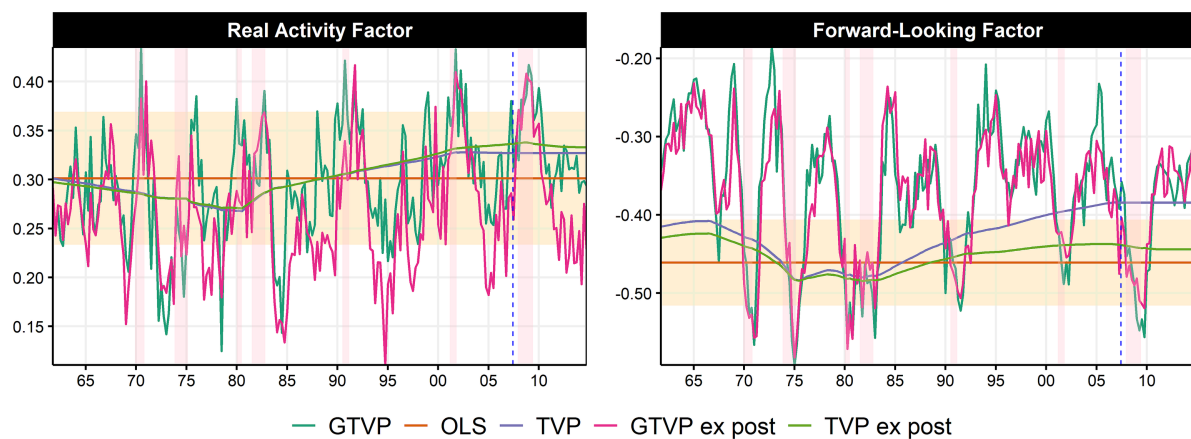


FIGURE 4 Unemployment rate (UR) equation β_t 's obtained with different techniques. Persistence is defined as $\phi_{1,t} + \phi_{2,t}$. TVPs estimated with a ridge regression as in Goulet Coulombe (2020a) and the parameter volatility is tuned with k-fold cross-validation—see Figure A18a in the supporting information (Appendix A8 in the supporting information) for a case where TVP parameter volatility is forced to be higher. Ex-post TVP means using the full sample for estimation and tuning as opposed to only using pre-2002 data as for GTVPs. The pale orange region is the OLS coefficient $\pm$ one standard error. Pink shading corresponds to NBER recessions.

provides a clumsy average of two regimes. In Figure 3, the obvious consequence of OLS' rigidity is being overresponsive to leading indicators during tranquil economic times and underresponsive when it matters. Section 4.3 will investigate formally the underlying variables driving this time variation.

4.2 | Comparing generalized TVPs with random walk TVPs

The relationship between random walk TVPs and GTVPs was evoked earlier. I compare them for the small factor model. I estimate standard TVPs using the two-step ridge regression technique developed by Goulet Coulombe (2020a).⁵ Conveniently, the procedure incorporates a cross-validation step that determines the optimal level of time variation in the random walks and a second step that allows in a data-driven way for some parameters to vary more and some other less. When forcing for more time variation against cross-validation's advice, results from the ridge regression are very similar (but overall less erratic) to what was obtained using a typical Kalman filter approach (the R function `tvreg`) or kernel smoothing (from the R package `tvReg`).

As Figure 3 suggested for $\gamma_{2,t}$, parameters can be subject to recurrent, rapid, and statistically meaningful shifts. Such behavior creates difficulties for random walk TVPs, which put the accent on smooth and slow structural change. Figure 4 confirms this conjecture. Standard TVPs look for long-run change when regime-switching behavior is the main driving force. As a result, they are flat and within OLS confidence bands, as often reported in the literature (D'Agostino et al., 2013). Of course, more action will mechanically be obtained for TVPs when considering a smaller amount of smoothness than what cross-validation proposed. In Appendix A8 in the supporting information, Figure A18a shows a similar exercise but using the optimal smoothing parameters (as picked by cross-validation) divided by 1000. This provides much more volatile random walk TVPs that are inclined, at certain specific moments, to follow the GTVPs. However, it is clear in Figure 4 that the end-of-sample/revision problem is worsened by the forced lack of smoothing.

It is known in the traditional TVP literature that there is a balance between flexible (but often erratic) β_t paths and very smooth ones where time variation may simply vanish. Since random walk TVPs are unfit for many forms of the time variation present in macroeconomic data, high bias estimates are usually reported as only they can keep variance at a manageable level. This can have serious implications. Relying too much on time-smoothing can create a mirage of long-run change and/or dissimulate parameters that mostly (but not solely) vary according to expansions/recessions.

Another concern, particularly consequential for forecasting, is the boundary problem. As discussed earlier, random walk TVP models forecasts can suffer greatly from it because by construction, forecasts are always made at the boundary of the variable on which the kernel is based—that is, time. One can deploy a one-sided kernel, but this only alleviate a

⁵In that paper, it is shown with simulations that this easier approach performs similarly well (and sometimes better) to traditional Bayesian TVP-VAR, for model sizes that the latter is able to estimate.

few pressing symptoms without attacking the heart of the problem. In sharp contrast, GTVPs use a large information set S_t to create the kernel, which implies that the likelihood of making a forecast at the boundary is rather low, unless the RF part constantly selects t as splitting variable.

Figure 4 shows for both random walk and generalized TVPs, their full sample versions (up to the end of 2014, “**ex-post**”) and their version with a training sample ending in 2007Q2 and projected on the test set data from 2017 to 2014 (the dashed blue line, “**ex ante**”). Mathematically, for $t \in [2007Q2, 2014Q4]$, $\hat{\beta}_t^{\text{ex ante}} = \hat{F}_{2007Q2}(S_t)$ and $\hat{\beta}_t^{\text{ex ante}} = \hat{F}_{2014Q4}(S_t)$ where the subscript on $\hat{F}$ means “estimated up to”. There are two main observations. First, GTVPs are much less prompt to rewrite recent history than random walk TVPs. Indeed, the green line and the magenta one closely follow each other all the way up to the end of the training sample. Second, while GTVPs can change many quarters after 2007Q2 (like the GDP constant), they are generally very close to each other at the boundary—especially when the time variation is statistically meaningful (like that of $\gamma_{F_2,t}$), which is what matters for forecasting. This is much less true of random walk TVPs as there are clear examples where the two versions differ for a long period of time (e.g., the intercept and the coefficient on F_2 in the GDP equation), and this often culminates at the boundary.

4.2.1 | Why and when MRF can fail to deliver better forecasts

MRF can sometimes be outperformed by simpler alternatives, like standard RF that incorporate MAFs. When that occurs, it is usually due to the inadequacy of the linear part rather than GTVPs themselves. Unlike traditional TVPs, GTVPs rarely provide a model worse than OLS.

Trivially, β_t helps understanding relative performance. For instance, in the case of forecasting inflation with the *quarterly* data set, ARRF does not supplant RF-MAF. The critical difference between ARRF (reported in Figure 5a) and its restricted analog is that the two autoregressive coefficients of the former are shut to 0. In Figure 5a, the estimates of ARRF broadly agree with the view that inflation persistence has substantially decreased during and following Volker disinflation (Cogley & Sargent, 2001).

In terms of anticipated forecasting performance, such decline in persistence suggests that a constrained version simply including μ_t may do better. The OOS evaluation period corresponds to the region of Figure 5a where $\phi_{1,t} + \phi_{2,t}$ is the nearest to 0. Given that observation, RF-MAF mildly improving upon ARRF is less surprising. An analogous finding emerges for GDP at many horizons. ARRF does not outperform RF-MAF like FA-ARRF and larger VARs versions of MRF do. GTVPs showcased in Figure 5b provide a simple explanation. There is only a limited role for persistence when allowing for a forest-driven μ_t . $\phi_{1,t} + \phi_{2,t}$ is below the OLS counterpart most of the time, and the credible 68% credible region frequently includes 0. The ensuing forecast is essentially a time-varying constant, which is what RF-MAF does. In sum, unlike many ML offerings, MRF successes and failures can be understood via a time-varying parameter interpretation. The helpfulness of this attribute cannot be overstated when thinking about future model improvements.

4.3 | Cutting down the forest, one tree at a time

GTVPs may be the result of an opaque ensemble of trees, but they are made out of observables or t . That is, they change but according to a *fixed* structure. Hence, the reduced-form coefficients could easily change and yet remain completely predetermined as long as $\mathcal{F}$ itself is stable. In this paradigm, a changing β_t is not necessarily empirical evidence supporting the Lucas' critique—rather, a changing $\mathcal{F}$ could be. Hence, dissecting $\mathcal{F}$ is inherently interesting. One way to get started on this is to use well-established measures of VI, originally proposed for RF in Breiman (2001). Those extract features drive the *prediction*. Conveniently, they can be adapted to inquire β_t . Then, one can capitalize on VI's insights to build interpretable small trees parsimoniously approximating $\beta_{t,k}$'s path.

The construction of upcoming graphs consists in two steps. I start by computing three different VI measures: VI_{OoB} (out-of-bag predictive performance), VI_{OoS} (OOS predictive performance) and VI_β (for a specific coefficient rather than the whole prediction). Essentially, any VI measure for a variable $S_{t,j}$ works by shuffling randomly variable j , recomputing (but not re-estimating) the prediction $\mathcal{F}(\tilde{S}_t)$ and then comparing it to real one $\mathcal{F}(S_t)$. Shuffling important variables will lead to a large gap between the outputs of $\mathcal{F}(\tilde{S}_t)$ and $\mathcal{F}(S_t)$. Appendix A5 in the supporting information contains a detailed explanation and a discussion on how the current approach relates to recent work in the ML interpretability literature. As a potential data set for the construction of a surrogate tree, I consider the union of the 20 most potent predictors as highlighted by any of the three VIs. The tree is pruned with a cost-complexity factor (usually referred to as cp) of 0.075. That tuning parameter is set such as to balance its capacity to mimic the original GTVP and potential for interpretation.

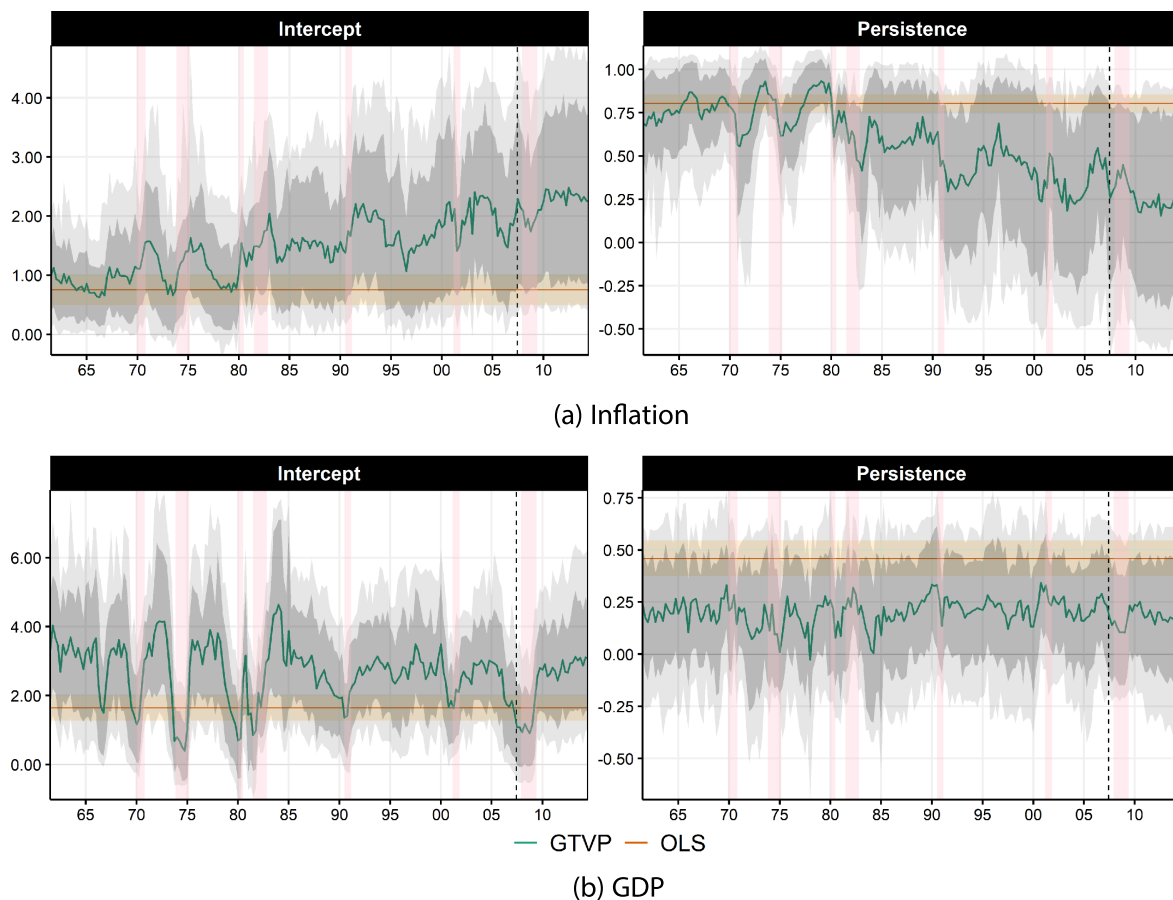


FIGURE 5 GTVPs of the one-quarter ahead forecasts using ARRF. Persistence is defined as $\phi_{1,t} + \phi_{2,t}$. The gray bands are the 32-68% and 10-90% quantiles of the tree distribution. The pale orange region is the OLS coefficient $\pm$ one standard error. The vertical dotted line is the end of the training sample. Pink shading is NBER recessions.

4.3.1 | Unemployment equation

I limit the attention μ_t and $\gamma_{2,t}$ paths, which were argued of greater importance to FA-ARRF's success in forecasting UR. Also, the nature of their variation is easier to characterize with a single tree (ex-post). Figure 6b,d shows that paths can sometimes be summarized succinctly using a handful of predictors.

Most of μ_t can be captured by two states which are determined by a cut-off on total private sector employees (USPRIV): 0.021 (increasing unemployment) and -0.018 (decreasing). This first layer basically classifies recession versus expansions in a very parsimonious way, which is inevitably crude and imperfect. The additional split on a MAF of nonfinancial leverage provides a more refined classification: There are more or less three states. The time series plot shows the alternation between two symmetrically opposed states of 0.021 and -0.025 (respectively, entering and exiting a recession) and a transitory (and seldomly visited) middle ground around 0.

The impact of F_2 on UR switches significantly, and most of the action can be summarized by a private sector employees dummy (USPRIV). The indicator's movement downwards—which usually commence from the *onset* of a recession—can double the effect of F_2 on UR in absolute terms. However, some high (absolute) $\gamma_{2,t}$ episodes would be left behind when merely using USPRIV. Those are retrieved by an additional split with a MAF of average corporate bonds yield with a BAA rating (lower medium grade).

The GTVP (green line) often plunges earlier than the ex-post surrogate tree's replica (orange). This is important, especially from a forecasting perspective. In Figure A15b in the supporting information (Appendix A8), it is clear that leading indicators (especially financial ones) play a prominent role in driving the GTVP $\gamma_{2,t}$ —well before USPRIV starts showing signs of an imminent downturn. Since F_2 is already composed mostly of forward-looking variables, this hints at a convex effect of market-based expectations proxies.

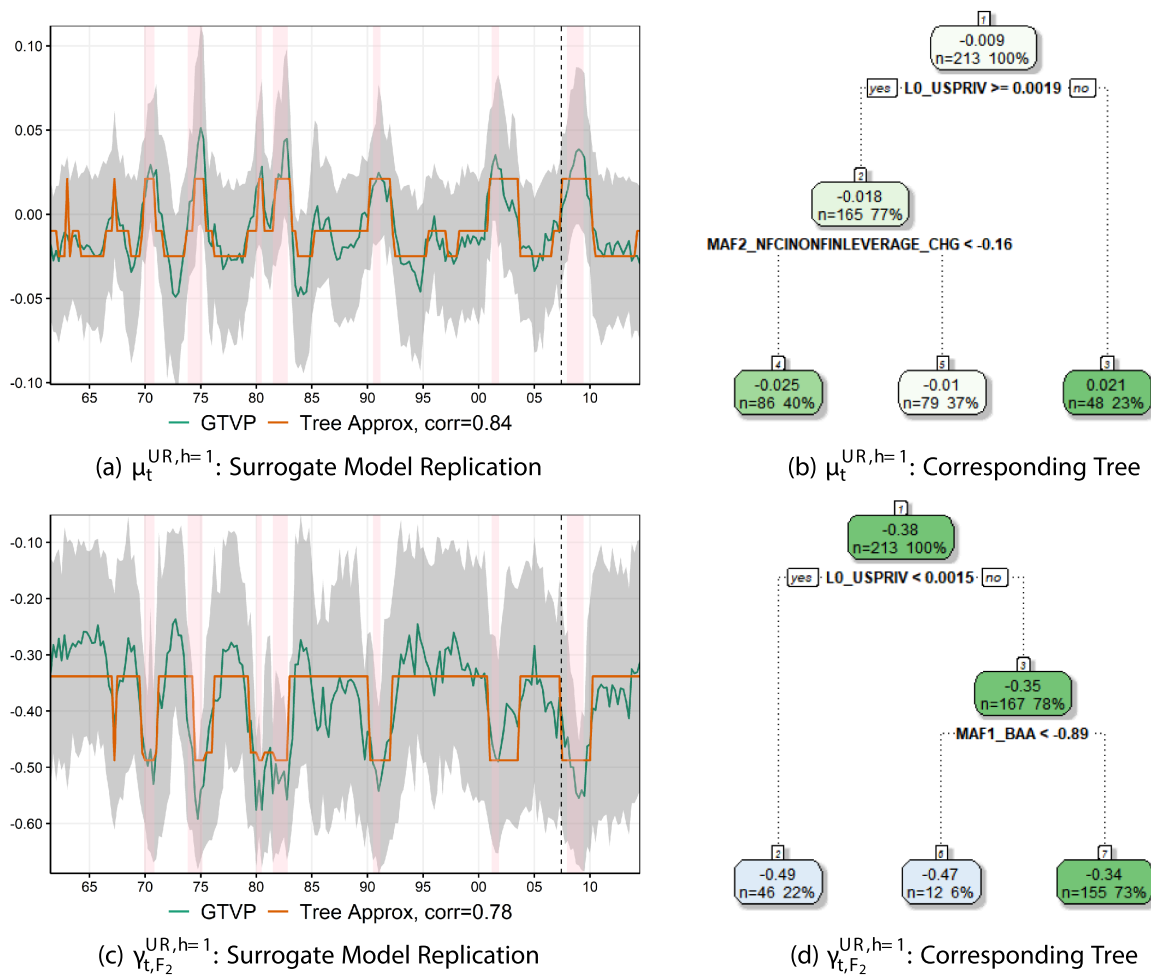


FIGURE 6 Surrogate $\beta_{t,k}$ trees. Shade is 32%–68% quantiles of tree distribution. Pink shading is NBER recessions.

4.3.2 | Monthly inflation equation

As detailed in Appendix A7 in the supporting information, FA-ARRF is a very competitive model for *monthly* inflation at all horizons. By its use of F_1 , the real activity factor, it has the familiar flavor of a PC. This is of interest given PCs have at best a very uneven forecasting track record (Atkeson & Ohanian, 2001; Stock & Watson, 2008). The curve could be time-varying in a way that annihilates its forecasting potential (Stock & Watson, 2008). Closely related, some have stipulated that the PC is nonlinear (Dolado et al., 2005; Lindé & Trabandt, 2019). Lastly, an adjacent point of view, which became increasing popular following the Great Recession, is that the PC had flattened to the point of predictive desuetude (Blanchard et al., 2015; Del Negro et al., 2020).

Of course, all those explanations amount to hypotheses on the nature of $\gamma_{1,t}$'s time variation of which MRF provides a very flexible account. It is worth emphasizing that MRF is estimated up to 2007Q2, unlike many of the above models explaining the “missing disinflation” after observing that it took place. The VI measures reported in Figure A16 in the supporting information showcase a “consensus” subset of variables that matters for inflation time variation. Three popular ones are the trend, MAF of building permits, and MAF of housing starts. The leading role for the trend suggests that exogenous time variation is important to explain inflation—which is well-known (Cogley & Sargent, 2001). Studying β_t -specific VIs suggest that this is mostly a feature of the intercept and persistence.

Figure 7 allows to reconcile PC forecasting evidence. For instance, a visible flat PC zone spans all of the 1990s, which constitutes most of the sample used in Atkeson and Ohanian (2001). It also includes the post-2008 period, which motivated Blanchard et al. (2015)'s inquiry. Most interestingly, for the latter era, $\gamma_{1,t}$ is predicted to head toward 0 OOS.

By looking at predictive performance results ex-post, Stock and Watson (2008) report that PC forecasts can outperform univariate benchmarks but only around turning points, which are detected ex-post. Accordingly, they note that the finding

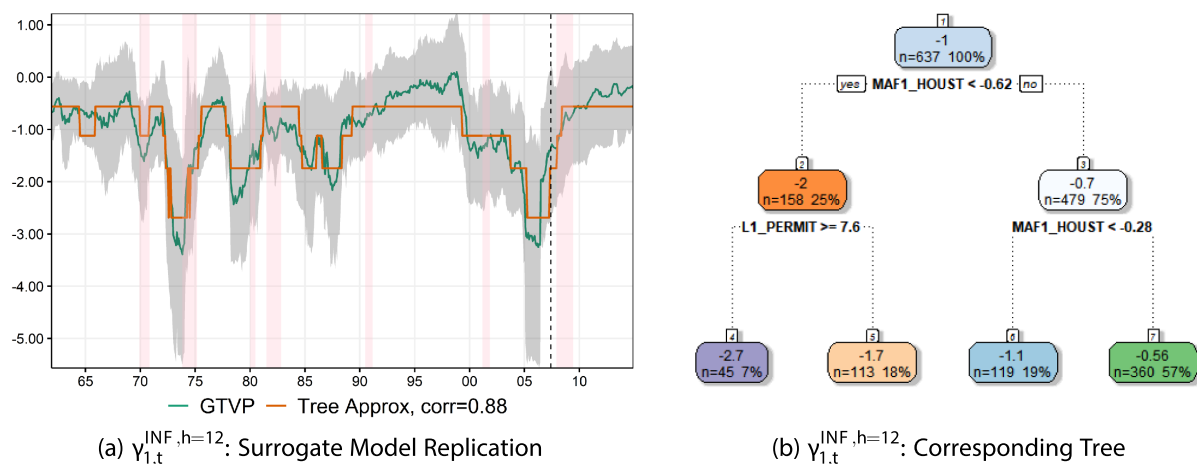


FIGURE 7 Surrogate $\beta_{t,k}$ trees for inflation. Shade is 32%–68% quantiles of the tree distribution. Pink shading is NBER recessions.

“cannot yet be used to improve forecasts”. More recently, Kotchoni et al. (2019) reinforce this view by showing that an ARMA(1,1) does fine at predicting inflation *except* in recessionary periods, where a data-rich environment can be helpful. But to capitalize on this, one needs a recession/expansion forecast.

MRF recognize this potential and relies on leading indicators of the housing market to activate $\gamma_{1,t}$ in a timely manner. This is particularly evident from looking at $\gamma_{1,t}$'s VI measure in Figure A16 in the supporting information and its resulting GTVP in Figure 7. Overall, we see that the relationship between inflation and economic activity is episodic, as conjectured by Stock and Watson (2008) and often prevails before recessions (but not all). Figure 7 suggests that inflation responds to the real activity factor when the housing market is booming. FA-ARRF differs from usual approaches by *not* using housing permits/starts as a replacement and/or additional “leading” output gap proxy (Stock & Watson, 2007). Rather, its role is to increase the curvature when the time is right. In that function, housing starts and permits are proxying for future economic activity, proposing a resolution to the conundrum posed by Stock and Watson (2008).

The predictive PC studied here differs in many aspects from those studied, for instance, in Blanchard et al. (2015). Importantly, F_1 summarizes mostly variables in first differences (or growth rates). A typical gap measure, being a deviation from a trend, will be much more persistent. Also, it remains negative for many years following a downturn. In contrast, F_1 , which is strongly correlated with UR change, will go back up as soon as UR stops growing.

4.4 | The PC: Not dead yet?

The behavior of inflation since the Great Recession—starting with the missing disinflation and followed by “missing inflation” of recent years—sparked renewed interest in the Phillips curve. Much attention has been given to its hypothesized flattening (Blanchard et al., 2015; Del Negro et al., 2020; Galí & Gambetti, 2019). This body of work supports the view that the PC coefficient (either reduced-form or semistructural) has substantially declined over the last decades. The focus on slow structural change is operationalized by the modeling strategy—either random walk TVPs or sample splitting at a specific date. Coibion and Gorodnichenko (2015) show less worry about PC's health. They rationalize post-2008 inflation with a simple OLS PC where expectations are based on consumer survey data rather than lags or professional forecasters. Del Negro et al. (2015) demonstrate that a standard DSGE (which encompasses a structural New Keynesian PC) is not baffled by post-2008 inflation since it relies on model-based forward-looking expectations of future marginal cost. More recently, Lindé and Trabandt (2019) articulate a theory supporting a nonlinear specification for the reduced-form PC, which could also account for the inflation puzzles punctuating the last 12 years. Nalewaik (2016) procures empirical support for the “convexity” hypothesis through a Markov-switching model, which, however, excludes by construction the possibility of long-run change in coefficients—as put forward by the aforementioned papers. Given this background and forecasting results reported earlier, a traditional PC must be a fertile ground for MRF.

I contribute to the literature by fitting an MRF which linear part corresponds to an expectation-augmented PC. X_t is inspired by what Blanchard et al. (2015) (henceforth BCS) consider:

$$\pi_t = \theta_t \hat{\pi}_t^{LR} + (1 - \theta_t) \hat{\pi}_t^{SR} + \phi_t u_t^{GAP} + \psi_t \pi_t^{IMP} + \epsilon_t, \quad (5)$$

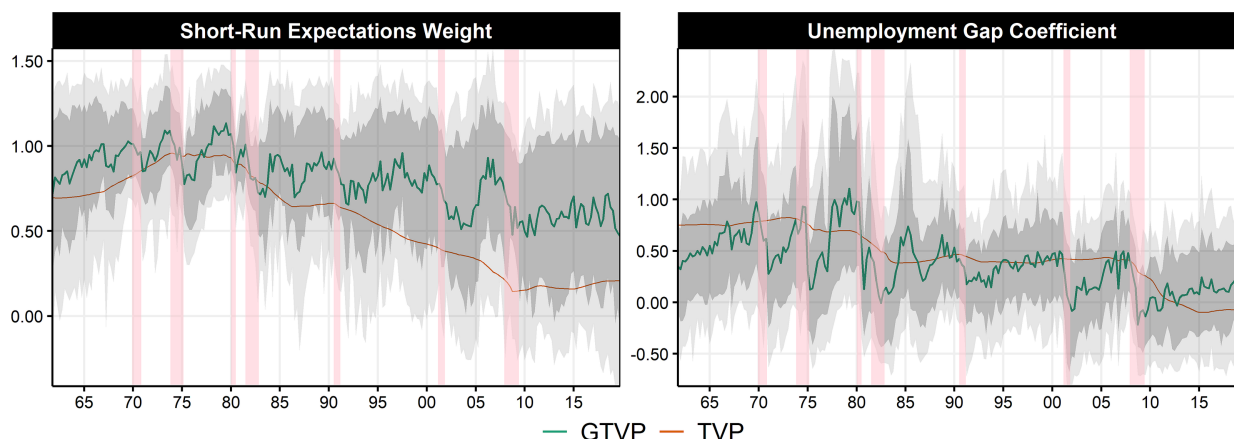


FIGURE 8 The gray bands are the 32%–68% and 10%–90% quantiles of the tree distribution. Pink shading is NBER recessions.

where π_t stands for CPI inflation, $\hat{\pi}_t^{LR}$ and $\hat{\pi}_t^{SR}$, respectively, for long-run and short-run inflation expectations. u_t^{GAP} represents the (negative) unemployment gap, and π_t^{IMP} is import prices inflation. I translate this to the MRF framework by making $\mu_t = \theta_t \hat{\pi}_t^{LR}$ the time-varying intercept, letting $\beta_{1,1} = 1 - \theta_t$ and by obtaining u_t^{GAP} by means of Hodrick–Prescott filtering. As in BCS, $\hat{\pi}_t^{SR}$ is the average inflation over the last four quarters. Hence, the estimated equation

$$\pi_t = \mu_t + \beta_{1,t} \hat{\pi}_t^{SR} + \beta_{2,t} u_t^{GAP} + \beta_{3,t} \pi_t^{IMP} + \varepsilon_t \quad (6)$$

does not impose the constraint implied by θ_t in Equation (5). However, estimation results will desirably have $\beta_{1,t} \in [0, 1]$ at almost any point in time. S_t is the same as that considered in the forecasting section. The data set runs up to 2019Q4.

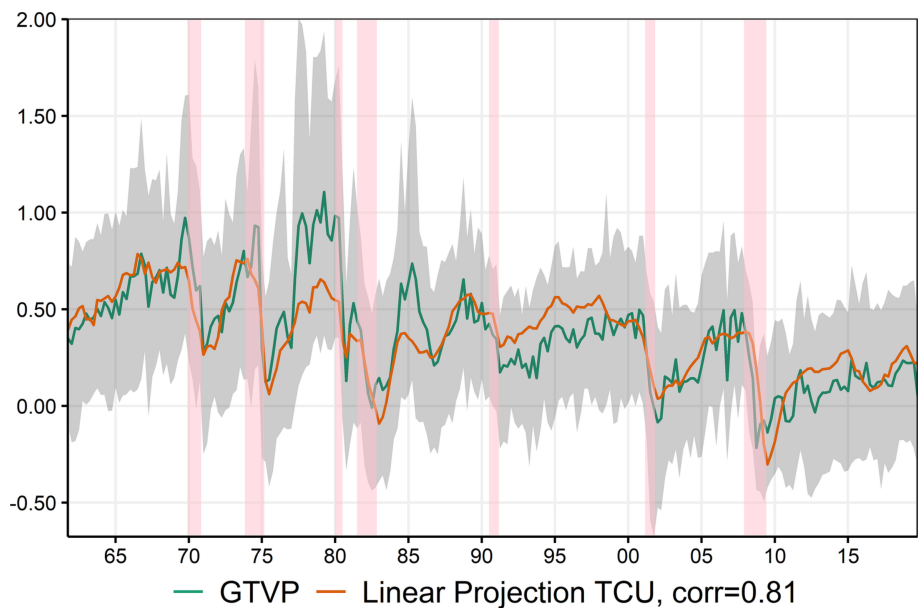
Figure 8 reports GTVPs of interest: the weight on short-run expectations and the output gap coefficient. Additionally, it contains traditional TVP estimates as means of comparison. The latter convey the usual wisdom: Inflation expectations slowly start to be more anchored from the mid-1980s. Around the same time, the unemployment/inflation trade-off begins its slow collapse. The updated data show that the TVP-based PC has further flattened to plain 0 in the last decade.

For $\beta_{1,t}$, the weight on short-run expectations, both methods agree that it has been decreasing steadily after the 1983 recession. But GTVPs highlight an additional pattern for the importance of $\hat{\pi}_t^{SR}$: It tends to increase during economic expansions, collapse during recessions, then start increasing again until the next downturn. Note that the phenomenon is also observed in Figure 5b for the simpler ARRF on quarterly inflation. The decrease in the coefficient (usually of about 0.25) is observed for every recession and usually lasts for some additional quarters after the end of it. The linear rise in the coefficient occurs for all expansions except those preceding the early 1990s and 2000s recessions, where the pattern is punctuated with additional peaks and troughs. The increased importance of short-run expectations with the age of the expansion is also observed for recent expansionary periods. Hence, the phenomenon is not merely a matter of the 1970s and the 1980s recessions being preceded by a sharp acceleration of inflation.

From a more statistical point of view, the sharp decline in $\beta_{1,t}$ following every recession suggests that in the aftermath of an important downward shock, the long-run inflation expectation is a more reliable predictor as it is minimally affected by recent events. As the expansion slowly progresses (and recessionary data points get out of the short-run average), $\hat{\pi}_t^{SR}$ becomes a more up-to-date and reliable barometer of future inflation conditions. This narrative is corroborated by VI (Figure A19 in the supporting information, Appendix A8) for $\beta_{1,t}$, which highlights not only the importance of the trend but also the recent lags of inflation.

When it comes to the low-frequency movement of the unemployment gap coefficient, both methods agree about a significant decline starting from the 1980s. However, GTVPs uncover additional heterogeneity. **First** and most strikingly, $\beta_{2,t}$ gets very close to 0 following every recession. This suggests a nonlinear Philipps' curve where inflation responds strongly to a very positive u_t^{GAP} but not so much to a negative one. **Second**, the 1970s and early 1980s are characterized as a series of peaks (preceding the first three recessions of the sample) rather than a sustained high coefficient. Traditional TVPs, by excessive time-smoothing, dissimulate the effects of inflationary spirals on $\beta_{2,t}$. Such prerecession accelerations still occur during the Great Moderation but in a much milder way. **Third**, VI measures (in Figure A19 in the supporting information) confirm the importance of activity indicators (like total capacity utilization [TCU]) in driving $\beta_{2,t}$ itself. The

FIGURE 9 The gray band is the 32%–68% quantiles of the tree distribution. Pink shading is NBER recessions.



correlation between $\beta_{2,t}$ and TCU is 0.81, and the correspondence between the two variables is striking in Figure 9. Many notable increases in $\beta_{2,t}$ are nicely matched (between the two 1970s recessions and before 2008). Of course, this simple characterization remains imperfect since it misses some highs (like the end of the 1970s) and predicts a higher $\beta_{2,t}$ in the years following the 2008–2009 recession. Generally, given the strong co-cyclical relationship between TCU and u_t^{GAP} , this is evidence of a *convex* PC.

The collapse of $\beta_{2,t}$ following recessions is not unique to 2008: It happened following *every* recession since 1960. As a result, inflation will rise when the economy is running well above its potential but has a much more muted response to economic slack. Recently, Lindé and Trabandt (2019) have shown that such a phenomenon can be rationalized by a New Keynesian DSGE model. Indeed, by allowing for additional strategic complementarity in firms price- and wage-setting behavior and solving the nonlinear model (rather than considering the linear approximation around the steady state), the authors obtain a state-dependent PC which becomes very flat during large downturns. This can explain both the small coefficient during recessions and its subsequent timid increase.

5 | CONCLUSION

I proposed a new time series model that **(i)** expands multiple nonlinear time series models, **(ii)** adapts Random Forest for macro forecasting, and **(iii)** can be interpreted as GTVPs. On the empirical front, the methodology provides substantial empirical gains over RF and competing nonlinear time series models. The resulting GTVPs have a very distinct behavior vis-à-vis standard random walk parameters. For instance, they adapt nicely to regime-switching behavior that seems pervasive for unemployment—while not neglecting potential long-run change. This finding is facilitated by the fact that MRF's byproducts can be visualized with standard time series plots, and a dispersion metric is provided by a variant of the Bayesian bootstrap. When looking at PCs in general, MRF finds both structural change in the persistence and regime-dependent behavior in the unemployment/inflation trade-off. In particular, a recurrent theme across all specifications is that the slowly decaying curve is also much steeper when the economy is overheating—in line with the convexity/nonlinearity hypothesis.

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OPEN RESEARCH BADGES



This article has been awarded Open Data Badge for making publicly available the digitally-shareable data necessary to reproduce the reported results. Data is available at DOI: 10.15456/jae.2023321.0115060814.

DATA AVAILABILITY STATEMENT

The data is publicly available at the Federal Reserve of St-Louis's web site.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of the article.

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